DO NOW — RECALL

Lesson 086 · Debugging Unfamiliar Algorithms

Name: _____ Class: _____ Date: _____

1. What is a loop invariant?

2. What is a boundary test?

3. What return value commonly indicates a failed indexed search?

4. In array [2,4,6], set a[1]=a[0]+a[2]. Give the final array.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Property maintained after each iteration.

2. Model answer

A test at or next to a threshold.

3. Model answer

−1.

4. Model answer

[2,8,6].

By the end, you can…

• Trace before changing code: list variables, loop indexes, condition and expected return.

• An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

• Correct code by explaining the cause and showing a fresh test, not just copying a working version.

Vocabulary — know these terms

logic error — Program runs but produces wrong answer.

boundary — Edge of valid range.

trace — Record changing states.

debugging — Finding and correcting faults.

NEW LEARNING

Read, annotate and explain

Observe

Trace before changing code: list variables, loop indexes, condition and expected return.

Explain

An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

Evaluate

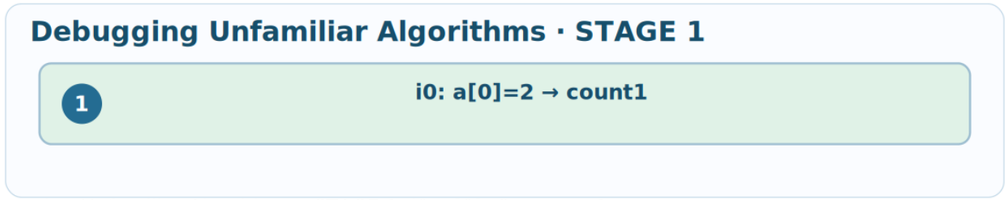
Correct code by explaining the cause and showing a fresh test, not just copying a working version.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Debugging Unfamiliar Algorithms · STAGE 1; i0: a[0]=2 → count1

New Learning

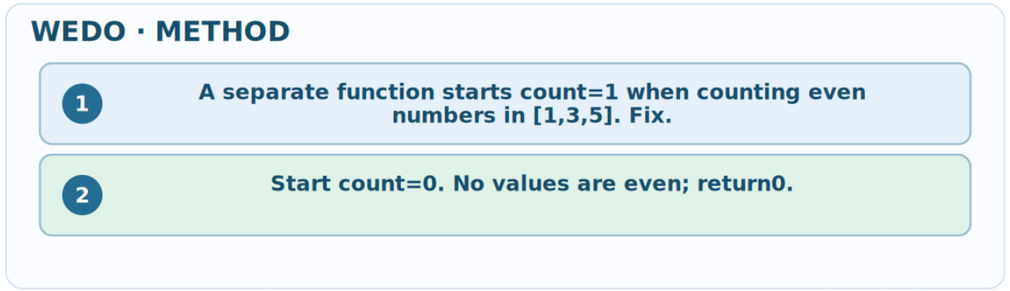
Debugging Unfamiliar Algorithms · STAGE 2; i0: a[0]=2 → count1; i1: a[1]=5 → count1

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A separate function starts count=1 when counting even numbers in [1,3,5]. Fix.

Initialising to1 incorrectly counts an item before any check.



WEDO · METHOD; A separate function starts count=1 when counting even numbers in [1,3,5]. Fix.; Start count=0. No values are even; return0.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

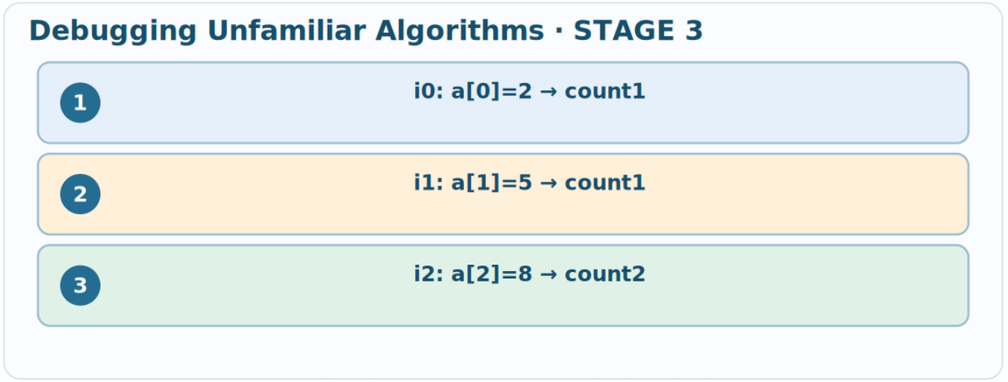
Guided 1 — CHECK

Model response
Initialising to1 incorrectly counts an item before any check.

New Learning · original lesson diagrams

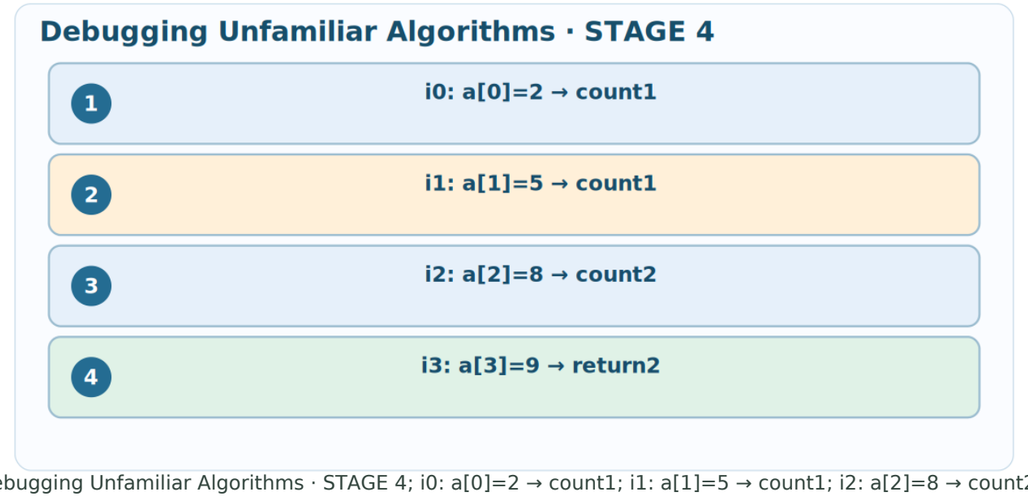
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Debugging Unfamiliar Algorithms · STAGE 3; i0: a[0]=2 → count1; i1: a[1]=5 → count1; i2: a[2]=8 → count2

New Learning



Debugging Unfamiliar Algorithms · STAGE 4; i0: a[0]=2 → count1; i1: a[1]=5 → count1; i2: a[2]=8 → count2; i3: a[3]=9 → return2

I DO — WORKED EXAMPLE

Follow the worked method

Repair for-loop condition i<=a.Length in an array scan.

Model response

For length4, valid indexes0–3. Checking a[4] would throw.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A loop using i<=Length can safely read a[i] for every i.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

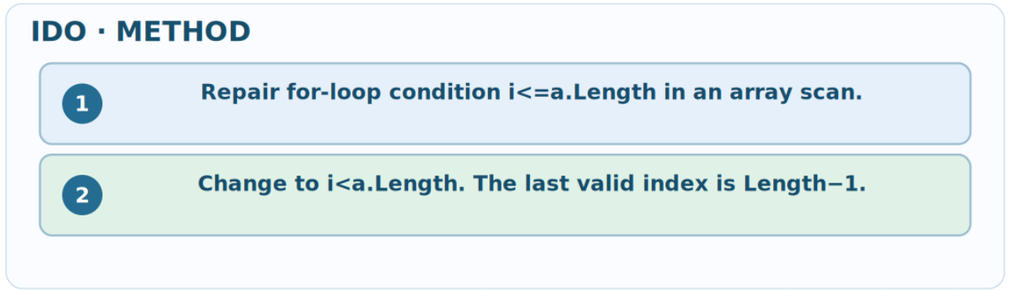
Develop four distinct tests for a function counting even values, with expected answers and reasons. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Repair for-loop condition i<=a.Length in an array scan.

For length4, valid indexes0–3. Checking a[4] would throw.



IDO · METHOD; Repair for-loop condition i<=a.Length in an array scan.; Change to i<a.Length. The last valid index is Length−1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A separate function starts count=1 when counting even numbers in [1,3,5]. Fix.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace CountEven([2,5,8,9]) with count after each item. [6]

Question 2

Explain why i<=a.Length is unsafe in a loop indexing a[i], and repair. [6]